

Probabilistic Benchmark and Transition-Width Ablation

Subject: the reviewer's two open requests, implemented and measured
Repository: ameye/fuzzy-audiogram **Reference runs:** archive/transition_2.0/, archive/transition_10.0/ **Verdict:** The reviewer's circularity objection is confirmed more strongly than they stated it. The requested widening is not a trade-off — it improves agreement, with a bootstrap interval excluding zero.

1. Why these two things

Two items were left open after the reproducibility work. Both were raised by the external review.

The comparator formulation. The reviewers noted that the machine-learning comparators regressed PTA-4 from seven frequency thresholds, four of which are the arithmetic terms of the target. That is a reconstruction task, not a clinical benchmark, and the near-perfect scores demonstrated it. They proposed a probabilistic multi-class model instead, benchmarked against the fuzzy membership vectors with Brier scores and cross-entropy.

The transition width. The reviewers asked for the graded region to be at least as wide as the ± 5 dB test-retest variability the partition represents, rather than the 2 dB margin in use. The partition construction was changed to satisfy this, but the *consequence* had not been measured.

2. What was implemented

2.1 The probabilistic benchmark

tests/probabilistic_benchmark.py. The fuzzy system's graded output is the six Ruspini memberships evaluated at the primary threshold. In this implementation the primary threshold is PTA-4 itself, so those memberships are a function of PTA-4 alone. They are non-negative and sum to one, so they are a probability distribution over severity and can be scored as one.

Against that, six probabilistic models were fitted on the training split and scored on the held-out test split:

Model	Features
Proportional-odds ordinal logistic	PTA-4
Multinomial logistic	PTA-4
Proportional-odds ordinal logistic	seven frequencies

Model	Features
Multinomial logistic	seven frequencies
Gradient boosting	seven frequencies
Random forest	seven frequencies

Metrics: quadratic-weighted kappa and accuracy from the arg-max label, multi-class Brier score, log loss, and expected calibration error on the top class.

Two feature sets are used deliberately. The fuzzy system consumes PTA-4 alone, so the PTA-only row is the like-for-like comparison that isolates the inference layer. The seven-frequency rows show what becomes available with the full audiogram, and they contain the four thresholds that define the reference label.

2.2 The transition-width ablation

`scripts/run_transition_ablation.sh`, `tests/transition_ablation_stats.py`. The same Ruspini construction and the same fixed calibration were run twice, differing only in the floor placed on the core gap:

- **2 dB arm** — the raw percentile gaps, which after construction range from 5.7 to 13.6 dB
- **10 dB arm** — gaps floored at 10 dB, giving 10.0 to 13.6 dB

Everything else was held constant, so the difference is attributable to the width of the graded region alone. The transition width is now settable through `FA_MIN_TRANSITION`, which made the ablation possible without editing code.

3. The circularity objection, confirmed

The result is decisive and it is stronger than the reviewers' own statement.

Model	Features	kappa	accuracy	Brier	log loss	ECE
Ordinal logistic	PTA-4	1.0000	100.0%	0.0000	0.0000	0.0000
Ordinal logistic	7 freq	1.0000	100.0%	0.0000	0.0000	0.0000
Multinomial logistic	PTA-4	0.9941	99.6%	0.0158	0.0335	0.0242
Multinomial logistic	7 freq	0.9934	99.6%	0.0107	0.0227	0.0135
Random forest	7 freq	0.9701	98.3%	0.0321	0.0599	0.0291

Model	Features	kappa	accuracy	Brier	log loss	ECE
FAI membership vector	PTA-4	0.9564	97.2%	0.0474	0.0805	0.0300
Gradient boosting	7 freq	0.9177	97.4%	0.0479	0.2771	0.0304
Uniform reference	—	0.0000	86.9%	0.8333	1.7918	0.7027

A proportional-odds ordinal logistic regression on PTA-4 alone reproduces the reference grade exactly, in every one of 3,912 ears, with a Brier score of zero.

That is not a good model. It is proof that the label is a deterministic monotone function of the input. The WHO grade is by construction a thresholding of PTA-4, so a model with a monotone latent recovers it exactly and the probabilities collapse to 0 and 1. Perfect separation drives the coefficients to the boundary and the Brier score to zero.

The consequence is that **agreement metrics cannot discriminate between models on this task**. Any model capable of representing monotone thresholds will score near-perfectly, and the remaining differences measure how well each approximation fits a step function, not how well it predicts anything.

This sharpens the reviewers' criticism rather than answering it. They objected that the comparator was reconstructing its target; the correct statement is that *any* supervised model on this target is reconstructing it, including the probabilistic one they proposed. The fuzzy system is not worse at the task. It is not doing the same kind of task.

3.1 What the comparison can still show

Two things survive, and both are informative.

Discrimination among approximations. The models do not all saturate. The ordinal logistic represents monotone thresholds exactly and scores 1.0000. The random forest approximates them piecewise and scores 0.9701. Gradient boosting scores 0.9177 — *worse than the fuzzy memberships*, on seven-frequency input, with more capacity.

That ordering is not incidental. A label defined by thresholds on a single continuous quantity is exactly the structure a set of trapezoidal membership functions represents natively: the partition's breakpoints *are* thresholds on decibels. A tree ensemble has to approximate the same step function with axis aligned splits on noisy inputs, and it does that less well. The fuzzy system

therefore has the correct inductive bias for this label, and the comparison demonstrates it.

Probabilistic quality. Where the models do differ is in the quality of their graded output, and there the fuzzy system is fully competitive:

- Brier 0.0474 against 0.0479 for gradient boosting, on richer input
- Log loss 0.0805 against 0.2771 for gradient boosting
- Expected calibration error 0.0300 against 0.0304

The fuzzy memberships are better calibrated than gradient boosting and much better on log loss, from less input. The two metrics beyond the label are therefore not decoration: they are the only place the comparison carries information, which is precisely why the reviewers asked for them.

A note on the labelling step. The membership vector's own arg-max recovers the reference grade at kappa 0.9564, which is *better* than the 0.9456 obtained from the separately calibrated FAI-score thresholds. The graded representation is a slightly better classifier than the crisp labelling step layered on top of it. That had not been observed before, because only the calibrated label was ever reported.

4. The transition-width ablation

Metric	2 dB gaps	10 dB floor	Difference	95% CI
Weighted kappa	0.9266	0.9456	+0.0190	[+0.0108, +0.0272]
Overall accuracy	94.61%	96.47%	+1.87 pp	[+1.28, +2.48]
Borderline accuracy	79.48%	85.09%	+5.61 pp	[+2.78, +8.48]
Clear accuracy	98.08%	99.09%	+1.01 pp	—

Paired bootstrap over 4,000 resamples, $n = 3,912$ ears. All three intervals exclude zero.

Of the 152 ears (3.9%) that the two widths label differently, the narrow construction gets 24.3% right and the wide one 72.4%.

The reviewers' requested widening improves the agreement metrics; it does not trade against them. Borderline agreement gains 5.6 points, which is the quantity the review specifically probed.

4.1 A correction to my own earlier analysis

I had predicted the opposite — that widening the graded region would reduce borderline agreement with the crisp reference by construction, since more ears would fall in a partially-graded state where the reference is a hard boundary. I said so in an earlier report and proposed the manuscript present it as an expected cost.

That was wrong, and the measurement refutes it. The reason is that the label-threshold calibration is fitted after the partition is fixed: a wider graded region changes the shape of the defuzzified score in a way the calibration can exploit, aligning the FAI thresholds with the WHO boundaries more effectively. Borderline accuracy rose from 79.5% to 85.1%.

The manuscript should therefore state the widening as a measured improvement with a confidence interval, not as a trade-off to be excused.

4.2 What the two widths actually differ in

2 dB gaps		10 dB floor	
normal	[-0.5, 0.0, 16.4, 30.0]	normal	[-0.5, 0.0, 16.4, 30.0]
mild	[16.4, 30.0, 35.7, 45.0]	mild	[16.4, 30.0, 35.7, 45.0]
moderate	[35.7, 45.0, 54.3, 60.0]	moderate	[35.4, 45.4, 52.1, 62.1]
moderately_severe	[54.3, 60.0, 67.1, 75.0]	moderately_severe	[52.1, 62.1, 66.1, 76.1]
severe	[67.1, 75.0, 83.6, 94.8]	severe	[66.1, 76.1, 83.6, 94.8]
profound	[83.6, 94.8, 120.0, 120.5]	profound	[83.6, 94.8, 120.0, 120.5]

Both are valid partitions summing to one everywhere. They differ only in the width of the bands where the membership is shared, and the wide arm’s bands are all 10.0 dB or more against a floor of 5.7 dB in the narrow arm.

5. What this means for the manuscript

Four changes are warranted.

Report the transition width as a measured improvement. Replace any framing of the wider band as a trade-off with the ablation result and its interval.

Add the probabilistic comparison, and state what it proves. The ordinal logistic’s exact reconstruction is the clearest possible evidence that the reference label is a deterministic function of its input. Reporting it converts the reviewers’

objection from a weakness into a stated property of the task, and makes the case for why the fuzzy representation is appropriate to it.

Lead with the metrics that discriminate. Agreement is saturated by construction. Brier score, log loss and calibration error are where the comparison lives, and the fuzzy memberships are competitive or better there against richer input.

Report the membership arg-max. Kappa 0.9564 from the graded representation against 0.9456 from the calibrated label is a better headline and a more honest description of what the system produces.

6. Artefacts

Scripts `tests/probabilistic_benchmark.py` — the probabilistic comparison `tests/transition_ablation_stats.py` — paired bootstrap on the ablation `scripts/run_transition_ablation.sh` — the ablation driver `scripts/pipeline_participant.py` — `FA_MIN_TRANSITION` override added

Results `data/output_participant/probabilistic_benchmark.json` `data/output_participant/transition_ablation.json` `archive/transition_2.0/` and `archive/transition_10.0/` — full runs

Consistency checks applied The benchmark aborts unless its reconstructed test set aligns with the saved predictions. That check caught two real errors during development: a split drawn with `choice` where the pipeline uses `permutation`, and severity bounds written `exclusive` where the pipeline uses `inclusive` upper bounds. The pipeline's `who_grade` is now imported rather than reimplemented, so the boundaries cannot diverge again.